CAR SHARING DATA ANALYSIS REPORT WITH SQL QUERIES

Project Title: Car Sharing Data Analysis Using Google Sheets and SQLite
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Tools Used: Google Sheets, DB Browser for SQLite, SQL, Microsoft Word

GOOGLE DRIVE LINK: https://drive.google.com/drive/folders/1wXEbfN-ovWIErRlJ_1uppeFM0KYz_V0e?usp=drive_link

1. INTRODUCTION

This project focuses on analyzing a car-sharing dateset to understand the factors influencing car-sharing demand. The analysis examines demand patterns across different days, months, seasons, hours, weather conditions, and temperature categories.

Google Sheets was used for data cleaning and transformation, while DB Browser for SQLite was used to create the database, manage tables, and execute SQL queries.

The main objective was to extract useful information from the dateset and present findings that can support better decision-making in car-sharing operations.

2. DATASET DESCRIPTION

The original dateset contained **8,708 records and 11 columns**. These columns included:

a. ID
b. Timestamp
c. Season
d. Holiday
e. Working day
f. Weather
g. Temperature
h. Feels-like temperature
i. Humidity
j. Windspeed
k. Demand

The dateset contained missing values in some columns, particularly temperature, feels-like temperature, humidity, and wind-speed.

3. DATA CLEANING AND TRANSFORMATION

Google Sheets was used to clean and prepare the dateset.

The following operations were performed:

a. Missing values in the temperature and feels-like temperature columns were replaced with their respective average values.
b. A new `temp_category` column was created to classify temperatures:

i. Below 10: Cold
ii. 10 to 25: Mild
iii. Above 25: Hot

c. A `temp_code` column was created by combining temperature, feels-like temperature, and temperature category.

d. Weather conditions were assigned numerical codes:

i. Clear or partly cloudy: 1
ii. Mist: 2
iii. Light snow or rain: 3
iv. Heavy rain, ice pellets, snow and fog: 4

e. Hour, weekday name, and month name were extracted from the timestamp.

f. Separate lookup sheets were created for weather, temperature, and time.

g. Duplicate records were removed from the lookup sheets.

The prepared data was exported into CSV files for database creation.

4. DATABASE DESIGN

A SQLite database named **carsharing** was created using DB Browser for SQLite.

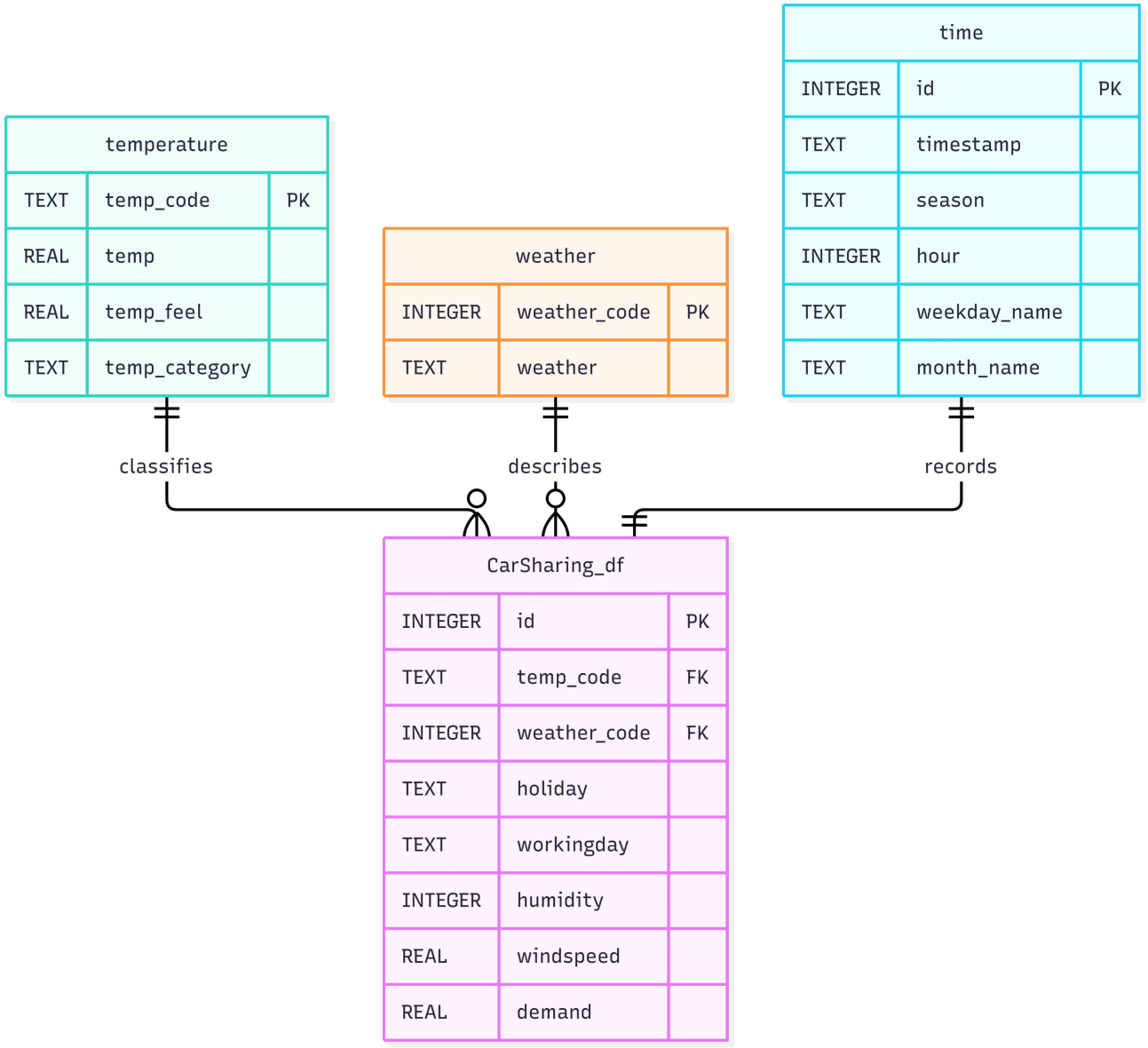
a. CarSharing_df: Main dateset containing demand and related variables.
b. CarSharing_df: Main dateset containing demand and related variables.
c. temperature: Temperature values and temperature categories.
d. weather: Weather descriptions and numerical weather codes.
e. time: Timestamp, season, hour, weekday, and month information.

The tables were connected using common fields:

f. temp_code connects CarSharing_df to temperature.
g. weather_code connects CarSharing_df to weather.
h. id connects CarSharing_df to time.

The relationships were checked using SQL joins.

Figure 1: Entity Relationship Diagram of the Car Sharing Database.



5. SQL QUERIES AND ANALYSIS

i. Highest Demand in 2017

```
SELECT timestamp, demand
FROM CarSharing_df
WHERE timestamp LIKE '2017%'
ORDER BY demand DESC
LIMIT 1;
```

Result:

- Date: 15 June 2017
- Time: 17:00 (5:00 PM)
- Demand: 6.4583

Interpretation: The highest recorded demand in 2017 occurred on 15 June at 5:00 PM.

ii. Average Demand by Weekday

```
SELECT weekday_name,
    ROUND(AVG(demand), 4) AS average_demand
FROM CarSharing_df
GROUP BY weekday_name
ORDER BY average_demand DESC;
```

Weekday	Average demand
Saturday	4.6019
Sunday	4.4974
Friday	4.4627
Tuesday	4.4268
Wednesday	4.4188
Monday	4.4078
Thursday	4.3532

Results:

Interpretation: Saturday recorded the highest average demand, while Thursday recorded the lowest.

iii. Average Demand by Month

```
SELECT month_name,
    ROUND(AVG(demand), 4) AS average_demand
FROM CarSharing_df
GROUP BY month_name
ORDER BY average_demand DESC;
```

Results:

Month	Average demand
July	4.9047
June	4.8925
May	4.7559
August	4.7287

October	4.5624
September	4.5501
April	4.4606
November	4.4395
December	4.2769
March	4.1900
February	3.9768
January	3.7820

Interpretation: July recorded the highest average demand, while January recorded the lowest.

iv. Average Demand by Season

```
SELECT season,
    ROUND(AVG(demand), 4) AS average_demand
FROM CarSharing_df
GROUP BY season
ORDER BY average_demand DESC;
```

Results:

Season	Average demand
Fall	4.7701
Summer	4.7033
Winter	4.4262
Spring	3.9842

Interpretation: Fall recorded the highest average demand, while Spring recorded the lowest.

v. Hourly Demand by Weekday

```
SELECT weekday_name,
    hour,
    ROUND(AVG(demand), 4) AS average_demand
FROM CarSharing_df
GROUP BY weekday_name, hour
ORDER BY weekday_name, hour;
```

Key findings:

a. Highest average hourly demand: Wednesday at 17:00, with 6.0817.
b. Lowest average hourly demand: Thursday at 04:00, with 1.1033.

Interpretation: Demand generally increased during busier daytime and evening periods, while early morning hours recorded lower demand.

vi. Demand by Weather Condition

```
SELECT w.weather,
      ROUND(AVG(c.demand), 4) AS average_demand,
      ROUND(MAX(c.demand), 4) AS highest_demand,
      ROUND(MIN(c.demand), 4) AS lowest_demand
FROM CarSharing_df c
JOIN weather w
ON c.weather_code = w.weather_code
GROUP BY w.weather
ORDER BY average_demand DESC;
```

Results:

Weather condition	Average demand	Highest demand	Lowest demand
Heavy rain/ice pellets/snow and fog	5.0999	5.0999	5.0999
Clear or partly cloudy	4.5440	6.7719	0.0000
Mist	4.4051	6.7274	0.0000
Light snow or rain	3.8394	6.7923	0.0000

Interpretation: The heavy-weather category recorded the highest average demand in the query results. However, its identical minimum, maximum, and average values should be treated cautiously because the number of records in that category was not independently verified.

vii. Monthly Humidity and Windspeed Statistics

```
SELECT
   month_name,
   ROUND(AVG(humidity), 2) AS average_humidity,
   ROUND(MAX(humidity), 2) AS highest_humidity,
   ROUND(MIN(humidity), 2) AS lowest_humidity,
   ROUND(AVG(windspeed), 2) AS average_windspeed,
   ROUND(MAX(windspeed), 2) AS highest_windspeed,
   ROUND(MIN(windspeed), 2) AS lowest_windspeed
FROM CarSharing_df
GROUP BY month_name
ORDER BY
   CASE month_name
      WHEN 'January' THEN 1
      WHEN 'February' THEN 2
      WHEN 'March' THEN 3
      WHEN 'April' THEN 4
```

```
    WHEN 'May' THEN 5
    WHEN 'June' THEN 6
    WHEN 'July' THEN 7
    WHEN 'August' THEN 8
    WHEN 'September' THEN 9
    WHEN 'October' THEN 10
    WHEN 'November' THEN 11
        WHEN 'December' THEN 12
    END;
```

Key findings:

a. Highest average humidity: September (74.84).
b. Lowest average humidity: April (55.66).
c. Highest average windspeed: April (15.59).
d. Lowest average windspeed: December (10.84).
e. Minimum windspeed across all months: 0.

Interpretation: September recorded the highest average humidity, while April recorded the highest average windspeed. December recorded the lowest average windspeed.

viii. Demand by Temperature Category

```
SELECT
   t.temp_category,
   ROUND(AVG(c.demand), 4) AS average_demand,
   ROUND(MAX(c.demand), 4) AS highest_demand,
   ROUND(MIN(c.demand), 4) AS lowest_demand
FROM CarSharing_df c
JOIN temperature t
ON c.temp_code = t.temp_code
GROUP BY t.temp_category
ORDER BY average_demand DESC;
```

Results:

Temperature category	Average demand	Highest demand	Lowest demand
Hot	4.9316	6.7923	0.0000
Mild	4.0903	6.7117	0.0000
Cold	3.3929	6.2166	0.0000

Interpretation: Hot conditions recorded the highest average demand, while cold conditions recorded the lowest average demand.

ix. Weather Analysis During the Highest-Demand Month

July was identified as the month with the highest average demand. The following query examined demand under different weather conditions during July.

```
SELECT
   w.weather,
   ROUND(AVG(c.demand), 4) AS average_demand,
   ROUND(MAX(c.demand), 4) AS highest_demand,
   ROUND(MIN(c.demand), 4) AS lowest_demand
FROM CarSharing_df c
JOIN weather w
ON c.weather_code = w.weather_code
WHERE c.month_name = 'July'
GROUP BY w.weather
ORDER BY average_demand DESC;
```

Results:

Weather condition	Average demand	Highest demand	Lowest demand
Mist	4.9302	6.7274	1.0986
Clear or partly cloudy	4.9249	6.7708	0.6931
Light snow or rain	4.4301	6.7081	0.0000

Interpretation: During July, mist recorded the highest average demand, closely followed by clear or partly cloudy conditions. Light snow or rain recorded the lowest average demand.

6. SUMMARY OF FINDINGS

The analysis produced the following major findings:

a. The highest recorded demand in 2017 occurred on 15 June at 5:00 PM.
b. Saturday had the highest average demand among weekdays.
c. July had the highest average monthly demand.
d. Fall recorded the highest average demand among seasons.
e. Wednesday at 5:00 PM recorded the highest average hourly demand.
f. September had the highest average humidity.
g. April had the highest average windspeed.
h. Hot temperature conditions recorded the highest average demand.
i. During July, mist recorded the highest average demand among the weather categories returned.

7. DATA QUALITY LIMITATION

During the final data review, an error was identified in the original `temp_code` values. The codes in the temperature lookup table were corrected, but the corresponding codes in the main table were not fully updated.

Therefore, the temperature-category analysis reflects the results obtained during the SQL analysis before that correction. The temperature relationship should be fully revalidated before treating the temperature findings as final. The other results should also be interpreted within the limitations of the supplied dataset.

8. CONCLUSION

This project demonstrated how spreadsheet tools and SQL can be used to clean, organise, store, and analyse a car-sharing dataset. The analysis revealed useful patterns in demand based on time, season, weather, humidity, windspeed, and temperature.

The findings indicate that demand varies across different periods and environmental conditions. Such information can help car-sharing operators understand usage patterns, plan vehicle availability, and make more informed operational decisions.

Overall, the project improved practical skills in data cleaning, database creation, SQL querying, data analysis, and report writing.